

Patrick (Pengcheng) Jiang

Ph.D. Student in Computer Science

University of Illinois at Urbana-Champaign

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Education

University of Illinois at Urbana-Champaign

Ph.D. in Computer Science

Champaign, USA

Aug. 2024 – Present

- GPA: 4.00/4.00 Advisors: Prof. Jiawei Han (thesis) and Prof. Jimeng Sun
- Research Focus: Information Retrieval, Knowledge Graphs, and LLMs for Healthcare

M.S. in Computer Science

Aug. 2022 – May 2024

- GPA: 3.97/4.00 Advisors: Prof. Jiawei Han and Prof. Jimeng Sun

Waseda University

B.E. in Computer Science and Engineering

Tokyo, Japan

Sep. 2017 – Sep. 2021

- GPA: 3.85/4.00 Advisors: Prof. Shinichi Honiden and Prof. Kenji Tei

Published Research Papers

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|-------------------|--|
| EMNLP'25 | Pengcheng Jiang , Xueqiang Xu, Jiacheng Lin, Jinfeng Xiao, Zifeng Wang, Jimeng Sun, and Jiawei Han, “s3: You Don’t Need That Much Data to Train a Search Agent via RL”. [pdf] [code] |
| COLM'25 | Pengcheng Jiang , Jiacheng Lin, Lang Cao, Runchu Tian, SeongKu Kang, Zifeng Wang, Jimeng Sun, and Jiawei Han, “DeepRetrieval: Hacking Real Search Engines and Retrievers with Large Language Models via Reinforcement Learning”. [pdf] [code] |
| ICLR'25 | Pengcheng Jiang , Cao Xiao, Minhao Jiang, Parminder Bhatia, Taha Kass-Hout, Jimeng Sun, and Jiawei Han, “Reasoning-Enhanced Healthcare Predictions with Knowledge Graph Community Retrieval”. [pdf] [code] |
| AAAI'25 | Pengcheng Jiang , Cao Xiao, Tianfan Fu, Parminder Bhatia, Taha Kass-Hout, Jimeng Sun, and Jiawei Han, “Bi-level Contrastive Learning for Knowledge-Enhanced Molecule Representations”. [pdf] [code] |
| NeurIPS'24 | Pengcheng Jiang , Lang Cao, Cao Xiao, Parminder Bhatia, Jimeng Sun, and Jiawei Han, “KG-FIT: Knowledge Graph Fine-Tuning Upon Open-World Knowledge”. [pdf] [code] |
| ICLR'24 | Pengcheng Jiang , Cao Xiao, Adam Cross, and Jimeng Sun, “GraphCare: Enhancing Healthcare Predictions with Personalized Knowledge Graphs”. [pdf] [code] |
| NAACL'24 | Pengcheng Jiang , Cao Xiao, Zifeng Wang, Parminder Bhatia, Jimeng Sun, and Jiawei Han, “TriSum: Learning Summarization Ability from Large Language Models”. [pdf] [code] |
| NAACL'24 | Pengcheng Jiang , Jiacheng Lin, Zifeng Wang, Jimeng Sun, and Jiawei Han, “GenRES: Rethinking Evaluation for Generative Relation Extraction in the Era of LLMs”. [pdf] [code] |
| ACL'23 | Pengcheng Jiang , Shivam Agarwal, Bowen Jin, Xuan Wang, Jimeng Sun, and Jiawei Han, “Text-Augmented Open Knowledge Graph Completion via Pre-Trained Language Models”. [pdf] [code] |
| EMNLP'24 | SeongKu Kang, Yunyi Zhang, Pengcheng Jiang , Dongha Lee, Jiawei Han, and Hwanjo Yu, “Taxonomy-guided Semantic Indexing for Academic Paper Search”. [pdf] [code] |
| SSCI'21 | Pengcheng Jiang , Kenji Tei, “OACAL: Finding Module-consistent Specifications to Secure Systems from Weakened User Obligations”. [pdf] |

Published Survey & Tool Papers

- KDD'25** **Pengcheng Jiang***, Siru Ouyang*, Yizhu Jiao, Ming Zhong, Runchu Tian, and Jiawei Han, “Retrieval And Structuring Augmented Generation with Large Language Models”. [\[pdf\]](#) (*survey paper*)
- KDD'25** Ran Xu, **Pengcheng Jiang**, Linhao Luo, Cao Xiao, Adam Cross, Shirui Pan, Jimeng Sun, and Carl Yang, “A Survey on Unifying Large Language Models and Knowledge Graphs for Biomedicine and Healthcare”. [\[pdf\]](#) (*survey paper*)
- KDD'23** Chaoqi Yang*, Zhengbang Wu*, **Patrick Jiang**, Zhen Lin, Junyi Gao, Benjamin Danek, and Jimeng Sun, “PyHealth: A Deep Learning Toolkit for Healthcare Predictive Modeling”. [\[pdf\]](#)[\[doc\]](#)[\[code\]](#)

Preprints & Tech Reports

- preprint'25** **Pengcheng Jiang**, Lang Cao, Ruike Zhu, Minhao Jiang, Yunyi Zhang, Jimeng Sun, and Jiawei Han, “RAS: Retrieval-And-Structuring for Knowledge-Intensive LLM Generation”. [\[pdf\]](#)[\[code\]](#)
- User Study** **Pengcheng Jiang**, Megan Lim, Adam Cross, and Jimeng Sun, “MedKG: Empowering Medical Education with Interactive Construction and Visualization of KGs via LLMs”, 2023. [\[pdf\]](#)[\[code\]](#)[\[software\]](#)

Professional Experience

Google Research

Student Researcher

- Research in information retrieval and reinforcement learning

Mountain View, USA

Aug. 2025 – Dec. 2025

Google Research

Student Researcher

- Research in information retrieval and reinforcement learning

Mountain View, USA

Jun. 2025 – Aug. 2025

GE HealthCare

AI Scientist Intern

- Developed KARE (ICLR'25), which constructs multi-sourced clinical concept knowledge graphs and integrates them with LLM reasoning for healthcare predictions

Seattle, USA

May 2024 – Aug. 2024

Ant Group

Software Engineer (AI Engineering)

- Developed Auto-Refit, a tool embedded in core MLOps for dynamic updates of deep learning models
- Enhanced engineering skills across the full ML development lifecycle

Hangzhou, China

Nov. 2021 – Jul. 2022

Academic Experience

University of Illinois at Urbana-Champaign

Research Assistant

- Working with Prof. Jiawei Han and Prof. Jimeng Sun on knowledge graphs, LLMs, and healthcare AI
- Published multiple papers in top-tier venues (NeurIPS, ICLR, EMNLP, NAACL, ACL, COLM, AAAI, KDD)

Champaign, USA

Sep. 2022 – Present

Teaching Assistant - CS598: Deep Learning for Healthcare

Jan. 2023 – May 2023

- Mentored students on deep learning models and guided them to build healthcare AI pipelines with PyHealth
- Graded project reports and presentations, provided one-on-one assistance on course material

Teaching Assistant - CS101: Intro Computing

Aug. 2022 – Dec. 2022

- Led lab sections and office hours, designed 200+ examination questions
- Taught fundamentals of data structures, algorithms, and programming (Python, MATLAB)

Massachusetts Institute of Technology

Research Projects with Prof. Mark Vogelsberger

- Developed a skin lesion diagnosis framework using convolutional neural networks

Cambridge, USA

May 2021 – Aug. 2021

Waseda University

Research Assistant

- Developed OACAL algorithm combining model checking and ML for automatic software specification revision

Tokyo, Japan

Jun. 2020 – Jul. 2021

Invited Talks & Tutorials

- 2025.08 Google Research Internal Talk, Google Research
- 2025.08 “MedKLM: Unifying Large Language Models and Knowledge Graphs for Medicine and Healthcare”, KDD Tutorial
- 2025.08 “Retrieval And Structuring Augmented Generation with Large Language Models”, KDD Tutorial
- 2025.06 General Research Talk, invited by Hong Kong University of Science and Technology (HKUST)
- 2025.05 “DeepRetrieval: Pushing the Boundaries of Information Retrieval with LLMs in Health Domain”, invited by National Institutes of Health (NIH)
- 2024.12 “Reasoning-Enhanced Healthcare Predictions with Knowledge Graphs”, invited by BioNLP Group at UMass Amherst
- 2023.08 “PyHealth Hands-on Tutorial”, KDD 2023 Tutorial

Professional Services

Paper Reviewer: NeurIPS, ICLR, ICML, EMNLP, NAACL, ACL, AAAI, AISTATS, KDD